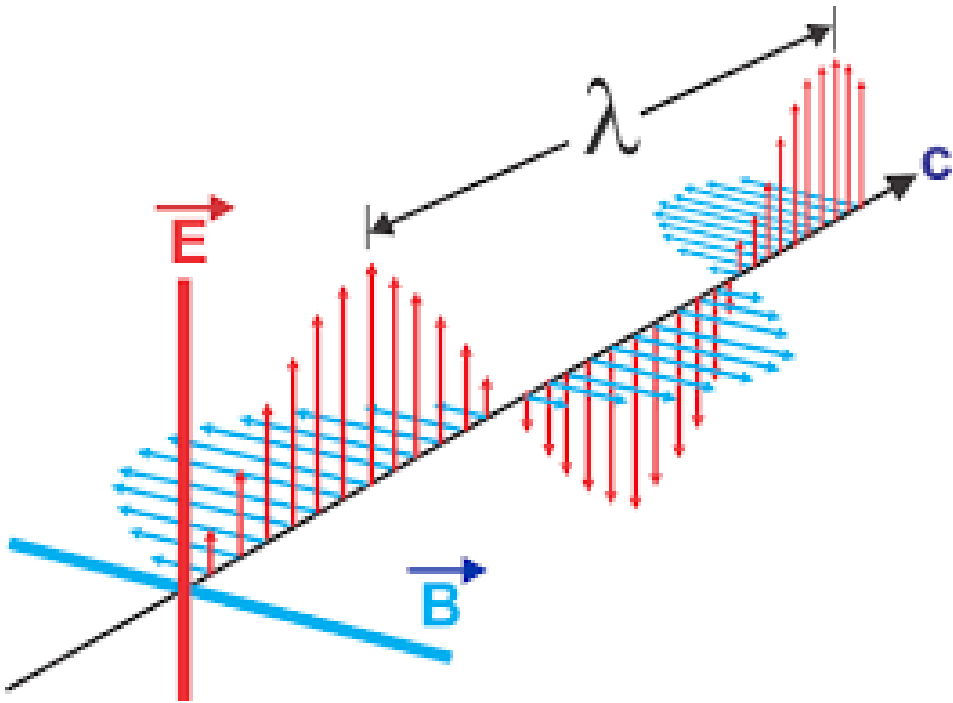


EM Wave & Waveguides

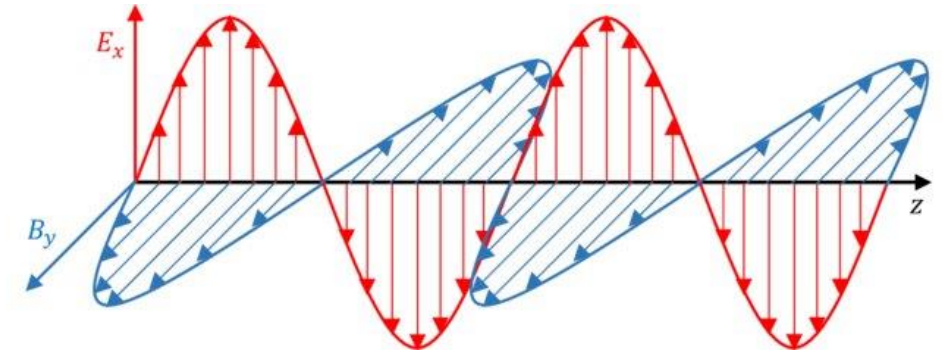
Part-3



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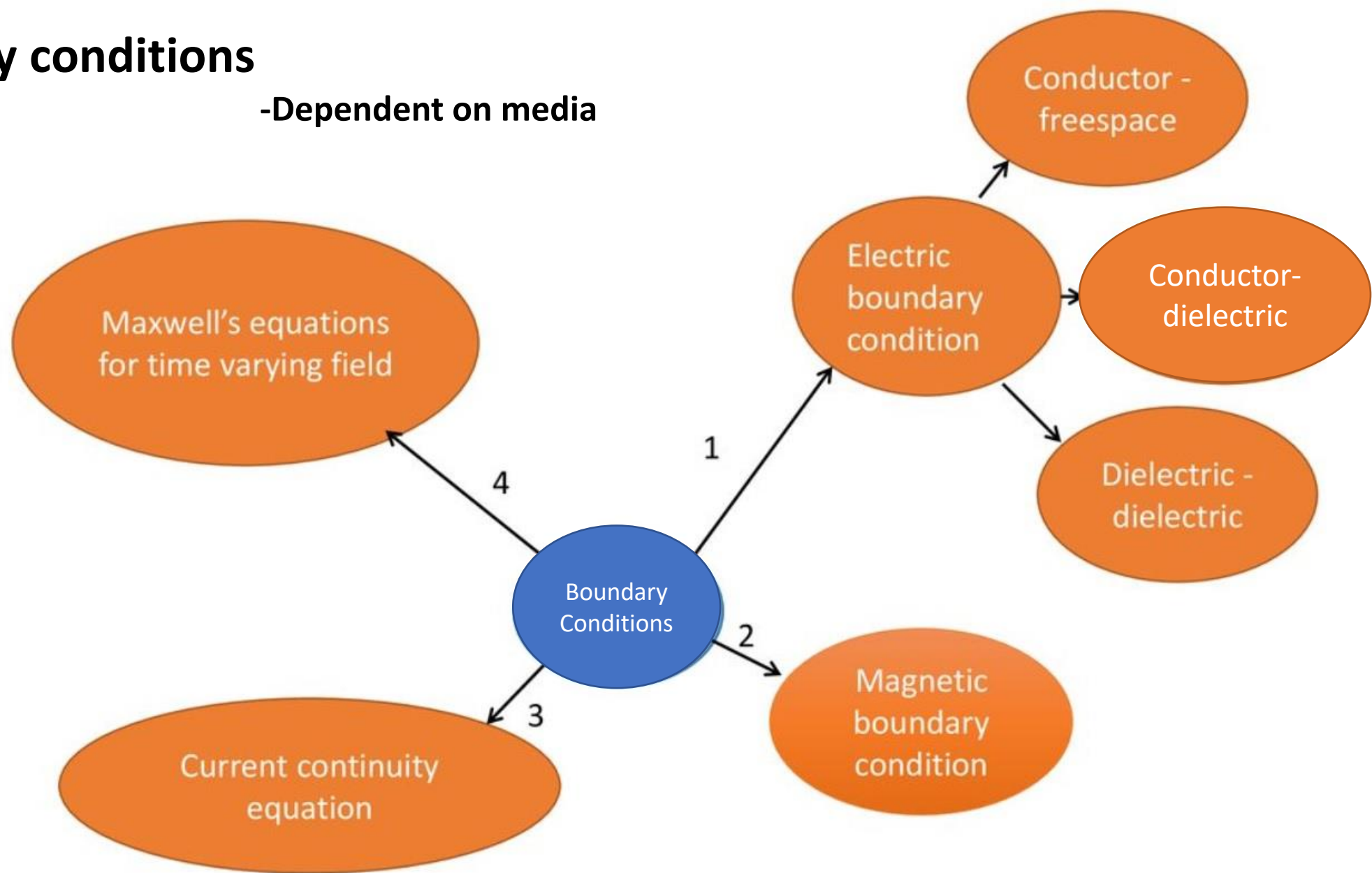
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Boundary conditions

-Dependent on media



In general, the fields $\mathbf{E}$, $\mathbf{B}$, $\mathbf{D}$, and $\mathbf{H}$ will be discontinuous at a boundary between two different media, or at a surface that carries charge density σ or current density $\mathbf{K}$.

$$\nabla \cdot \vec{D} = \rho_f$$



$$D_{1n} - D_{2n} = \rho_f$$

$$\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t},$$



$$E_{1t} = E_{2t},$$

$$\nabla \cdot \vec{B} = 0,$$



$$B_{1n} = B_{2n},$$

$$\nabla \times \vec{H} = \vec{J}_f + \frac{\partial \vec{D}}{\partial t},$$

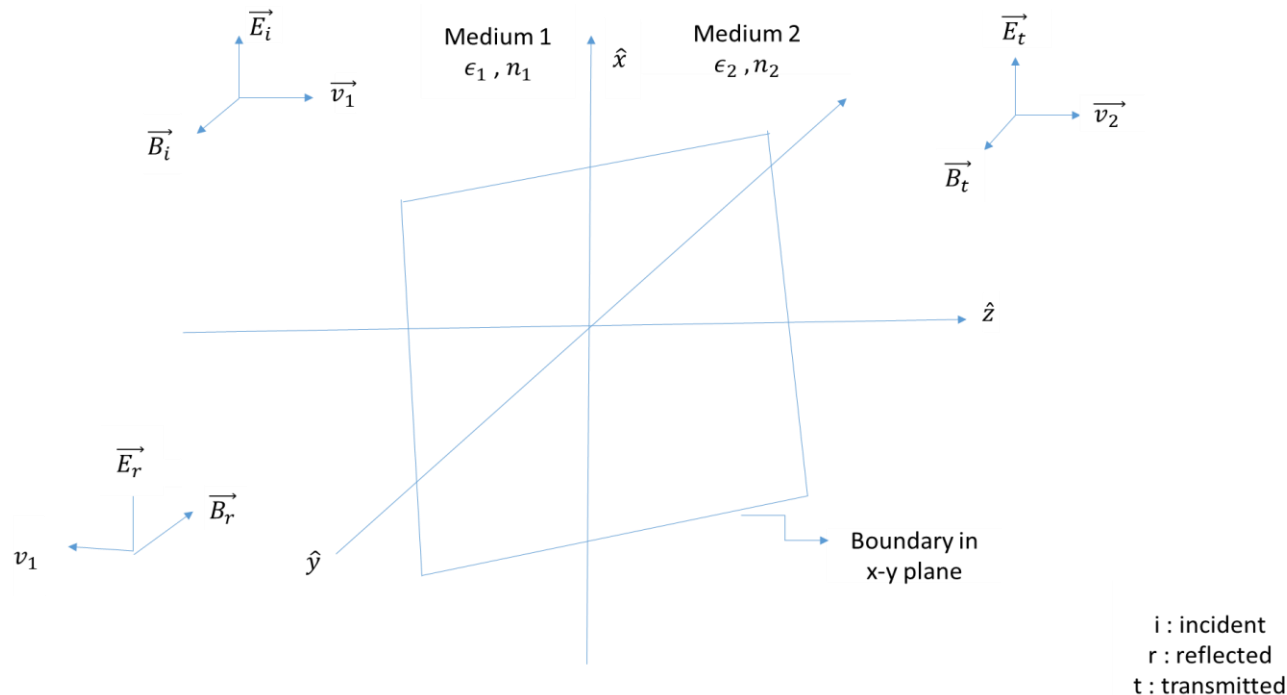


$$H_{1t} - H_{2t} = J_f$$

Boundary Conditions

What happens when an electromagnetic wave passes from one linear / homogeneous medium into another (e.g., vacuum, gas, air, water, oil)?

A monochromatic plane electromagnetic wave of frequency ω propagating in $+\hat{z}$ direction, which is linearly polarized in $+\hat{x}$ direction encounters a boundary between two linear media, lies in the x-y plane. Thus, this electromagnetic wave approaches the boundary from left and is at normal incidence to the boundary:



Boundary condition:1 The normal component of D is continuous across the interface at $z=0$:

$$D_1^\perp = D_2^\perp$$

$$\epsilon_1 E_1^\perp = \epsilon_2 E_2^\perp \quad \dots\dots\dots (1) \quad (\text{as } \vec{D} = \epsilon \vec{E})$$

Boundary condition:2 The tangential component of E is continuous across the interface at $z=0$:

$$E_1^\parallel = E_2^\parallel \quad \dots\dots\dots (2)$$

Boundary condition:3 The normal component of B is continuous across the interface at $z=0$:

$$B_1^\perp = B_2^\perp \quad \dots\dots\dots (3)$$

Boundary condition:4 The tangential component of H is continuous across the interface at $z=0$:

$$H_1^\parallel = H_2^\parallel \quad \dots\dots\dots (4) \quad \left(\frac{B_1^\parallel}{\mu_1} = \frac{B_2^\parallel}{\mu_2} \right)$$

Medium:1

Incident electromagnetic wave:

Propagates in $+\hat{z}$ – direction (i.e., $\hat{k}_i = +\hat{k}_1$) with polarization $+\hat{x}$

$$\text{Therefore, } \vec{E}_i(z, t) = \widetilde{E}_{0i} e^{i(k_1 z - \omega t)} \hat{x} \dots\dots\dots (5)$$

$$\vec{B}_i(z, t) = \frac{\hat{k}_i \times \vec{E}_i(z, t)}{v_1}$$

From $\hat{n} \times \vec{E}_0 = v \vec{B}_0$ where $\hat{n}$ ~unit vector in the propagation direction

$$\vec{B}_i(z, t) = \frac{\widetilde{E}_{0i} e^{i(k_1 z - \omega t)}}{v_1} \hat{y} \dots\dots\dots (6)$$

Reflected electromagnetic wave:

$$\vec{E}_r(z, t) = \widetilde{E}_{0r} e^{i(-k_1 z - \omega t)} \hat{x} \dots\dots\dots (7)$$

$$\vec{B}_r(z, t) = \frac{\hat{k}_r \times \vec{E}_r(z, t)}{v_1}$$

$$\vec{B}_r(z, t) = - \frac{\widetilde{E}_{0r} e^{i(-k_1 z - \omega t)}}{v_1} \hat{y} \dots\dots\dots (8)$$

Medium 2:

Transmitted electromagnetic wave:

$$\vec{\widetilde{E}}_t(z, t) = \widetilde{E}_{0t} e^{i(k_2 z - \omega t)} \hat{x} \quad \dots\dots\dots (9)$$

$$\vec{\widetilde{B}}_t(z, t) = \frac{\widehat{k}_t \times \vec{\widetilde{E}}_t(z, t)}{v_2}$$

$$\vec{\widetilde{B}}_t(z, t) = \frac{\widetilde{E}_{0t} e^{i(k_2 z - \omega t)}}{v_2} \hat{y} \quad \dots\dots\dots (10)$$

At the interface, i.e., at $z=0$ (in the x-y plane) the boundary conditions 1-4 must be satisfied for total E and B Fields present on either side of the interface between the two media:

Boundary condition 1: Normal D continuous at $z=0$: (refers to the x-y boundary i.e. in the $+\hat{z}$ direction)

$$\epsilon_1 E_{1tot}^\perp(z=0, t) = \epsilon_2 E_{2tot}^\perp(z=0, t) \quad \dots\dots\dots (11)$$

Boundary condition 2: Tangential E continuous at $z=0$: (refers to the x-y boundary i.e. in the x-y plane)

$$E_{1tot}^\parallel(z=0, t) = E_{2tot}^\parallel(z=0, t) \quad \dots\dots\dots (12)$$

Boundary condition 3:

$$B_{1tot}^\perp(z=0, t) = B_{2tot}^\perp(z=0, t) \quad \dots\dots\dots (13)$$

Boundary condition 4: Tangential H is continuous at $z=0$:

$$\frac{B_{1tot}^\parallel(z=0, t)}{\mu_1} = \frac{B_{2tot}^\parallel(z=0, t)}{\mu_2} \quad \dots\dots\dots (14)$$

For plane electromagnetic wave at normal incidence on the boundary at $z=0$ lying in the x-y plane, note that no components of E or B (incident, reflected or transmitted) are allowed to be along $\pm z$ directions because of the E and B fields transversality requirements $\rightarrow$ arising from the constraints imposed by Maxwell's equations.

Because of this boundary condition 1 & 3 impose no restriction here;

$$E_{1tot}^{\perp} = 0; E_{2tot}^{\perp} = 0$$

$$\text{And } B_{1tot}^{\perp} = 0; B_{2tot}^{\perp} = 0$$

Only restriction on plane electromagnetic wave propagating with normal incidence on the boundary at $z = 0$ are imposed by boundary condition 2 & 4.

In medium 1 i.e. $z \leq 0$:

$$\left. \begin{aligned} \overrightarrow{E_{1tot}^{\parallel}} &= \overrightarrow{E_i} + \overrightarrow{E_r} \\ \frac{\overrightarrow{B_{1tot}^{\parallel}}}{\mu_1} &= \frac{\overrightarrow{B_i}}{\mu_1} + \frac{\overrightarrow{B_r}}{\mu_1} \end{aligned} \right\} \dots\dots\dots (15)$$

In medium 2 ; $z \geq 0$:

$$\overrightarrow{E_{2tot}^{\parallel}} = \overrightarrow{E_t} \text{ and } \frac{\overrightarrow{B_{2tot}^{\parallel}}}{\mu_2} = \frac{\overrightarrow{B_t}}{\mu_2} \dots\dots\dots (16)$$

Then from boundary condition 2:

$$\overrightarrow{E_{1tot}^{\parallel}} \Big|_{z=0} = \overrightarrow{E_{2tot}^{\parallel}} \Big|_{z=0}$$

$$\vec{\widetilde{E}}_i (z=0,t) + \vec{\widetilde{E}}_r (z=0,t) = \vec{\widetilde{E}}_t (z=0,t) \quad \text{(from equation 15 \& 16)}$$

$$\text{Or, } \widetilde{E}_{0i} e^{-i\omega t} + \widetilde{E}_{0r} e^{-i\omega t} = \widetilde{E}_{0t} e^{-i\omega t} \quad \text{(using equation 5, 7 \& 8)}$$

$$\text{Or, } \widetilde{E}_{0i} + \widetilde{E}_{0r} = \widetilde{E}_{0t} \quad \dots\dots\dots (17)$$

Using boundary condition 4:

$$\left. \frac{\vec{\widetilde{B}}_{1tot}}{\mu_1} \right|_{z=0} = \left. \frac{\vec{\widetilde{B}}_{2tot}}{\mu_1} \right|_{z=0}$$

$$\frac{\vec{\widetilde{B}}_i (z=0,t)}{\mu_1} + \frac{\vec{\widetilde{B}}_r (z=0,t)}{\mu_1} = \frac{\vec{\widetilde{B}}_t (z=0,t)}{\mu_2} \quad \dots\dots\dots \text{(using equation 15)}$$

$$\frac{\widetilde{E}_{0i} e^{-i\omega t}}{\mu_1 v_1} - \frac{\widetilde{E}_{0r} e^{-i\omega t}}{\mu_1 v_1} = \frac{\widetilde{E}_{0t} e^{-i\omega t}}{\mu_2 v_2} \quad \dots\dots\dots \text{(using equation 6,8 \& 10)}$$

$$\frac{\widetilde{E}_{0i}}{\mu_1 v_1} - \frac{\widetilde{E}_{0r}}{\mu_1 v_1} = \frac{\widetilde{E}_{0t}}{\mu_2 v_2} \quad \dots\dots\dots (18)$$

Let us define $\beta = \frac{\mu_1 v_1}{\mu_2 v_2}$

Then, equation (18) becomes

$$\widetilde{E}_{0i} - \widetilde{E}_{0r} = \beta \widetilde{E}_{0t} \quad \dots\dots\dots (19)$$

From (17) & (19)

$$2\widetilde{E_{0i}} = (1 + \beta) \widetilde{E_{0t}}$$

$$\text{Or, } \widetilde{E_{0t}} = \frac{2}{(1+ \beta)} \widetilde{E_{0i}} \quad \text{..... (20)}$$

$$2\widetilde{E_{0r}} = (1 - \beta) \widetilde{E_{0t}}$$

$$\text{Or, } \widetilde{E_{0t}} = \frac{2}{(1- \beta)} \widetilde{E_{0r}} \quad \text{..... (21)}$$

Substituting equation (21) into (20)

$$\widetilde{E_{0r}} = \frac{(1- \beta)}{(1+ \beta)} \widetilde{E_{0i}}$$

$$\text{So, } \frac{\widetilde{E_{0r}}}{\widetilde{E_{0i}}} = \frac{(1- \beta)}{(1+ \beta)} \quad \text{and} \quad \frac{\widetilde{E_{0t}}}{\widetilde{E_{0i}}} = \frac{2}{(1+ \beta)} \quad \text{..... (22)}$$

If $\mu_1 = \mu_2 \approx \mu_0$, then

$$\beta = \frac{\mu_1 v_1}{\mu_2 v_2} = \frac{v_1}{v_2} = \frac{n_2}{n_1}$$

$$\text{then } \frac{\widetilde{E_{0r}}}{\widetilde{E_{0i}}} = \frac{v_2-v_1}{v_2+v_1} = \frac{n_1-n_2}{n_1+n_2}$$

$$\text{and } \frac{\widetilde{E_{0t}}}{\widetilde{E_{0i}}} = \frac{2v_2}{v_2+v_1} = \frac{2n_1}{n_1+n_2}$$

$$R = \left| \frac{\widetilde{E_{or}}}{\widetilde{E_{oi}}} \right|^2 = \left| \frac{n_1 - n_2}{n_1 + n_2} \right|^2$$

$$T = \left| \frac{\widetilde{E_{ot}}}{\widetilde{E_{oi}}} \right|^2 = \frac{4 n_1^2}{(n_1 + n_2)^2}$$

}

Waveguide

A waveguide is a structure that guides waves, such as electromagnetic waves or sound, with minimal loss of energy by restricting the transmission of energy to one direction.

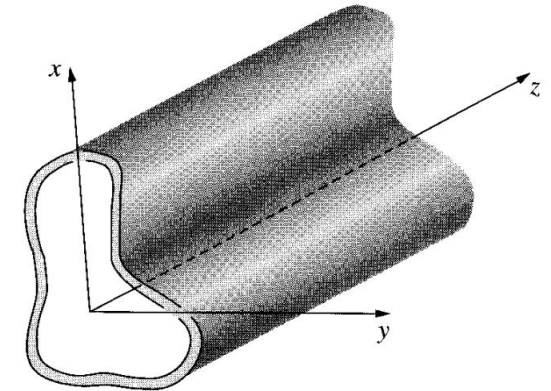
Consider EM waves confined to the interior of a hollow pipe, or waveguide.

Let us assume that the waveguide is a perfect conductor, so that

$$\vec{E} = 0 \text{ and } \vec{B} = 0 \text{ (Inside the material itself)}$$

The boundary conditions at the inner wall;

$$E_{\parallel} = 0, B^{\perp} = 0 \dots \dots \dots (1)$$



Reflection at a Conducting Surface

We are interested in monochromatic waves that propagate down the tube, so

$\vec{E}$ and $\vec{B}$ have the form

$$\vec{E}(x, y, z, t) = \vec{E}_0(x, y)e^{i(k_z z - \omega t)} \dots \dots \dots (2)$$

$$\vec{B}(x, y, z, t) = \vec{B}_0(x, y)e^{i(k_z z - \omega t)}$$

(Direction of propagation is along $+\hat{z}$ direction)

Generalized boundary conditions

In the interior of the waveguide, away from the walls, Maxwell's eqⁿs must be satisfied, which is for empty space or air with $\epsilon \approx \epsilon_0$ and $\mu = \mu_0$ are

$\vec{\nabla} \cdot \vec{E} = 0$	$\vec{\nabla} \cdot \vec{B} = 0$	 (3)
$\vec{\nabla} \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$	$\vec{\nabla} \times \vec{B} = \mu_0 \epsilon_0 \frac{\partial \vec{E}}{\partial t} = \frac{1}{c^2} \frac{\partial \vec{E}}{\partial t}$	

What restrictions arising from the boundary conditions in (1) are imposed on $\vec{E}$ and $\vec{B}$ in above eq (3) ?

Our $\vec{E}$ and $\vec{B}$ fields interior to the wave guide will be of the forms:

$$\vec{E}(x,y,z,t) = \vec{\widetilde{E}}_0(x,y)e^{i(k_z z - \omega t)}$$

With $\vec{\widetilde{E}}_0(x,y) = \widetilde{E}_{0x}(x,y)\hat{x} + \widetilde{E}_{0y}(x,y)\hat{y} + \widetilde{E}_{0z}(x,y)\hat{z}$ (4)

and $\vec{B}(x,y,z,t) = \vec{\widetilde{B}}_0(x,y)e^{i(k_z z - \omega t)}$

with $\vec{\widetilde{B}}_0(x,y) = \widetilde{B}_{0x}(x,y)\hat{x} + \widetilde{B}_{0y}(x,y)\hat{y} + \widetilde{B}_{0z}(x,y)\hat{z}$ (5)

Let us insert these expressions into Faraday's law and Ampere's law in eqⁿ (3), we get

Faraday's law:

$$\frac{\partial \widetilde{E}_{0y}}{\partial x} - \frac{\partial \widetilde{E}_{0x}}{\partial y} = i\omega \widetilde{B}_{0z} \dots \dots \dots (6)$$

$$\frac{\partial \widetilde{E}_{0z}}{\partial y} - \frac{\partial \widetilde{E}_{0y}}{\partial z} = i\omega \widetilde{B}_{0x} \dots \dots \dots (7)$$

Now $\frac{\partial \widetilde{E}_{0y}}{\partial z} = ik_z \widetilde{E}_{0y} \rightarrow$ because of z dependence of the factor $e^{i(k_z z - \omega t)}$ so Eq. (7) becomes

$$\frac{\partial \widetilde{E}_{0z}}{\partial y} - ik_z \widetilde{E}_{0y} = i\omega \widetilde{B}_{0x} \dots \dots \dots (8)$$

Finally,

$$\frac{\partial \widetilde{E}_{0x}}{\partial z} - \frac{\partial \widetilde{E}_{0z}}{\partial x} = i\omega \widetilde{B}_{0y} \dots \dots \dots (9)$$

Again as $\frac{\partial \widetilde{E}_{0x}}{\partial z} = ik_z \widetilde{E}_{0x}$, we have

$$ik_z \widetilde{E}_{0x} - \frac{\partial \widetilde{E}_{0z}}{\partial x} = i\omega \widetilde{B}_{0y} \dots \dots \dots (10)$$

Thus, we have

$$\frac{\partial \widetilde{E}_{0y}}{\partial x} - \frac{\partial \widetilde{E}_{0x}}{\partial y} = i\omega \widetilde{B}_{0z}$$

$$\frac{\partial \widetilde{E}_{0z}}{\partial y} - ik_z \widetilde{E}_{0y} = i\omega \widetilde{B}_{0x}$$

$$ik_z \widetilde{E}_{0x} - \frac{\partial \widetilde{E}_{0z}}{\partial x} = i\omega \widetilde{B}_{0y}$$

..... (a)

..... (b)

.....(c)

..... (11)

Ampere’s law

$$\frac{\partial \widetilde{B}_{0y}}{\partial x} - \frac{\partial \widetilde{B}_{0x}}{\partial y} = \frac{-i\omega}{c^2} \widetilde{E}_{0z} \dots\dots\dots (12)$$

$$\frac{\partial \widetilde{B}_{0z}}{\partial y} - \frac{\partial \widetilde{B}_{0y}}{\partial z} = \frac{-i\omega}{c^2} \widetilde{E}_{0x} \dots\dots\dots (12)$$

But $\frac{\partial \widetilde{B}_{0y}}{\partial z} = ik_z \widetilde{B}_{0y}$

$$\frac{\partial \widetilde{B}_{0z}}{\partial y} - ik_z \widetilde{B}_{0y} = \frac{-i\omega}{c^2} \widetilde{E}_{0x} \dots\dots\dots (14)$$

$$\frac{\partial \widetilde{B}_{0x}}{\partial z} - \frac{\partial \widetilde{B}_{0z}}{\partial x} = \frac{-i\omega}{c^2} \widetilde{E}_{0y} \dots\dots\dots (15),$$

$$ik_z \widetilde{B}_{0x} - \frac{\partial \widetilde{B}_{0z}}{\partial x} = \frac{-i\omega}{c^2} \widetilde{E}_{0y} \dots\dots\dots (16)$$

Thus, we have

$\frac{\partial \widetilde{B}_{0y}}{\partial x} - \frac{\partial \widetilde{B}_{0x}}{\partial y} = \frac{-i\omega}{c^2} \widetilde{E}_{0z}$	 (a)	
$\frac{\partial \widetilde{B}_{0z}}{\partial y} - ik_z \widetilde{B}_{0y} = \frac{-i\omega}{c^2} \widetilde{E}_{0x}$	 (b)	 (17)
$ik_z \widetilde{B}_{0x} - \frac{\partial \widetilde{B}_{0z}}{\partial x} = \frac{-i\omega}{c^2} \widetilde{E}_{0y}$	 (c)	

Using eqn 11 (c) and 17 (b), we get

$$ik_z \widetilde{E}_{0x} - \frac{\partial \widetilde{E}_{0z}}{\partial x} = i\omega \left(\frac{\partial \widetilde{B}_{0z}}{\partial y} + \frac{i\omega}{c^2} \widetilde{E}_{0x} \right) \frac{1}{ik_z}$$

Or

$$ik_z \widetilde{E}_{0x} - \frac{\partial \widetilde{E}_{0z}}{\partial x} = \frac{\omega}{k_z} \frac{\partial \widetilde{B}_{0z}}{\partial y} + \frac{i\omega^2}{k_z c^2} \widetilde{E}_{0x}$$

$$\widetilde{E}_{0x} = \frac{1}{ik_z - \frac{i\omega^2}{k_z c^2}} \left(\frac{\omega}{k_z} \frac{\partial \widetilde{B}_{0z}}{\partial y} + \frac{\partial \widetilde{E}_{0z}}{\partial x} \right)$$

$$\widetilde{E}_{0x} = \frac{1}{\frac{i}{k_z} \left(k_z^2 - \frac{\omega^2}{c^2} \right)} \left(\frac{\omega}{k_z} \frac{\partial \widetilde{B}_{0z}}{\partial y} + \frac{\partial \widetilde{E}_{0z}}{\partial x} \right)$$

$$\widetilde{E}_{0x} = \frac{ik_z}{\left(\frac{\omega^2}{c^2} - k_z^2 \right)} \left(\frac{\omega}{k_z} \frac{\partial \widetilde{B}_{0z}}{\partial y} + \frac{\partial \widetilde{E}_{0z}}{\partial x} \right)$$

$$\widetilde{E}_{0x} = \frac{i}{\left(\frac{\omega}{c} \right)^2 - k_z^2} \left(k_z \frac{\partial \widetilde{E}_{0z}}{\partial x} + \omega \frac{\partial \widetilde{B}_{0z}}{\partial y} \right) \dots \dots \dots (18)$$

Similarly, we can get

$$\widetilde{E}_{0y} = \frac{i}{\left(\frac{\omega}{c} \right)^2 - k_z^2} \left(k_z \frac{\partial \widetilde{E}_{0z}}{\partial y} - \omega \frac{\partial \widetilde{B}_{0z}}{\partial x} \right) \dots \dots \dots (19)$$

$$\widetilde{B}_{0x} = \frac{i}{\left(\frac{\omega}{c} \right)^2 - k_z^2} \left(k_z \frac{\partial \widetilde{B}_{0z}}{\partial x} - \omega \frac{\partial \widetilde{E}_{0z}}{\partial y} \right) \dots \dots \dots (20)$$

$$\widetilde{B}_{0y} = \frac{i}{\left(\frac{\omega}{c} \right)^2 - k_z^2} \left(k_z \frac{\partial \widetilde{B}_{0z}}{\partial y} + \omega \frac{\partial \widetilde{E}_{0z}}{\partial x} \right) \dots \dots \dots (21)$$

Now, we need to insert in 18-21 in remaining two Maxwell's equations

$$\frac{\partial \widetilde{E}_{0x}}{\partial x} + \frac{\partial \widetilde{E}_{0y}}{\partial y} + \frac{\partial \widetilde{E}_{0z}}{\partial z} = 0 \dots \dots \dots (22)$$

$$\frac{\partial \widetilde{B}_{0x}}{\partial x} + \frac{\partial \widetilde{B}_{0y}}{\partial y} + \frac{\partial \widetilde{B}_{0z}}{\partial z} = 0 \dots \dots \dots (23)$$

Substituting equation 18 and 19 in 22, we get

$$\frac{i}{\left(\frac{\omega}{c}\right)^2 - k_z^2} \left[k_z \frac{\partial^2 \widetilde{E}_{0z}}{\partial x^2} + \omega \frac{\partial^2 \widetilde{B}_{0z}}{\partial x \partial y} \right] + \frac{i}{\left(\frac{\omega}{c}\right)^2 - k_z^2} \left[k_z \frac{\partial^2 \widetilde{E}_{0z}}{\partial y^2} - \omega \frac{\partial^2 \widetilde{B}_{0z}}{\partial x \partial y} \right] + ik_z \widetilde{E}_{0z} = 0$$

$$\frac{ik_z}{\left(\frac{\omega}{c}\right)^2 - k_z^2} \left(\frac{\partial^2 \widetilde{E}_{0z}}{\partial x^2} + \frac{\partial^2 \widetilde{E}_{0z}}{\partial y^2} \right) + ik_z \widetilde{E}_{0z} = 0$$

$$\frac{\partial^2 \widetilde{E}_{0z}}{\partial x^2} + \frac{\partial^2 \widetilde{E}_{0z}}{\partial y^2} + ik_z \frac{\left(\frac{\omega}{c}\right)^2 - k_z^2}{ik_z} \widetilde{E}_{0z} = 0$$

$$\frac{\partial^2 \widetilde{E}_{0z}}{\partial x^2} + \frac{\partial^2 \widetilde{E}_{0z}}{\partial y^2} + \left\{ \left(\frac{\omega}{c}\right)^2 - k_z^2 \right\} \widetilde{E}_{0z} = 0$$

$$\left[\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \left(\frac{\omega}{c} \right)^2 - k_z^2 \right] \widetilde{E}_{0z} = 0 \dots \dots \dots (24)$$

Similarly, we get

$$\left[\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \left(\frac{\omega}{c} \right)^2 - k_z^2 \right] \widetilde{B}_{0z} = 0 \dots \dots \dots (25)$$

Case-1 Longitudinal component of $\vec{\widetilde{E}} = 0$

$\widetilde{E}_{0z} = 0$ but $\widetilde{B}_{0z} \neq 0 \rightarrow$ transverse electric (TE) waves

Case 2 Longitudinal component of $\vec{\widetilde{B}} = 0$

$\widetilde{B}_{0z} = 0$ but $\widetilde{E}_{0z} \neq 0 \rightarrow$ transverse magnetic (TM) waves

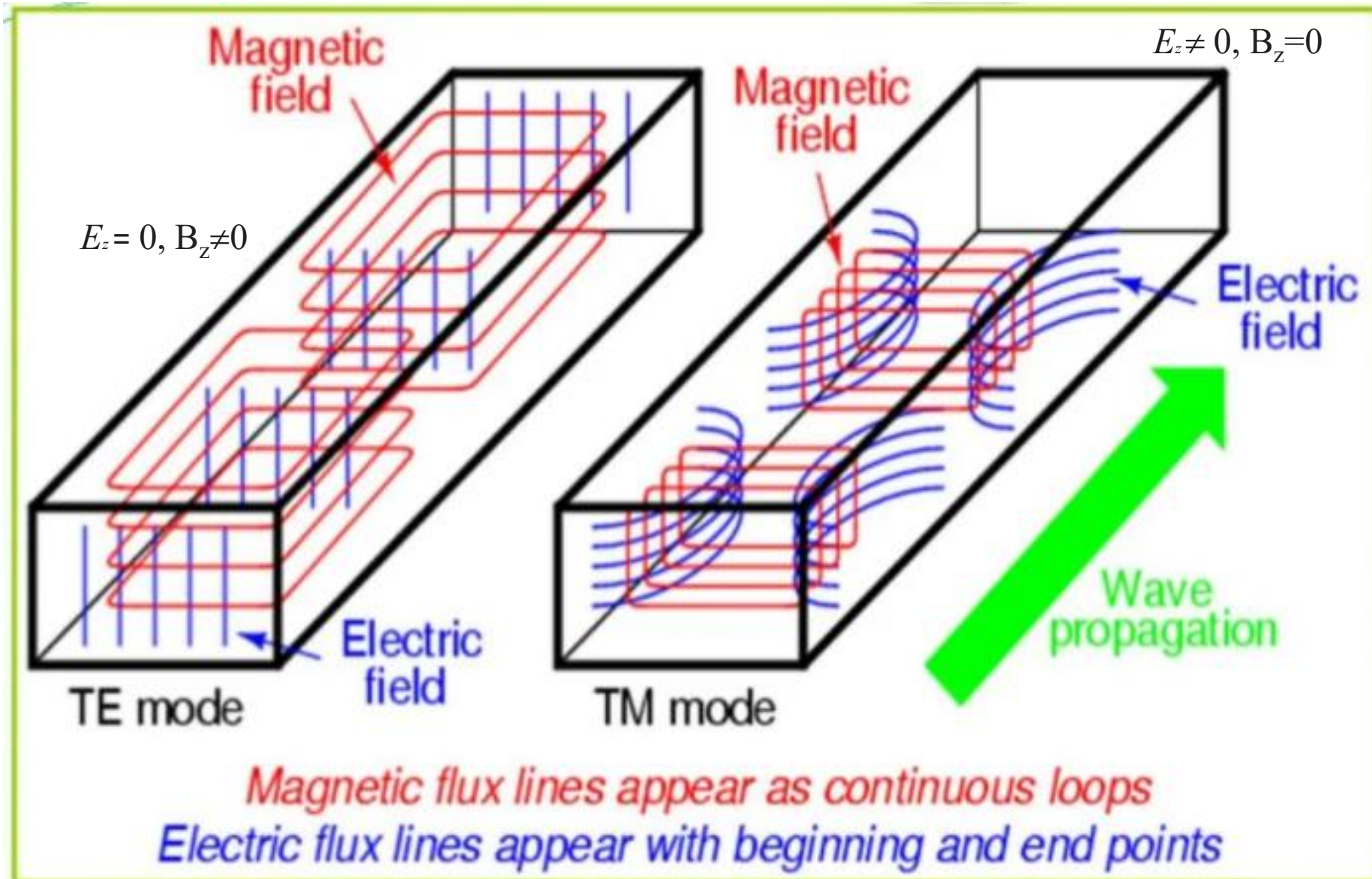
Case 3 Transverse electric and magnetic waves

Both $\widetilde{E}_{0z} = \widetilde{B}_{0z} = 0$

$\rightarrow$ no wave propagation, TEM Waves

Hence, transverse electric and magnetic waves cannot propagate in hollow waveguides.

Basic features of waveguide



Thanks